TT-S-179B
June 12, 1975
SUPERSEDING
Fed. Spec. TT-S-179A
June 7, 1966

FEDERAL SPECIFICATION

SEALER SURFACE: PIGMENTED OIL, FOR PLASTER AND WALLBOARD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an oil-type pigmented wall sealer for sizing interior walls of plaster and wallboard.

1.2 Classification.

1.2.1 The sealer shall be of one type and one grade as specified (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

SS-L-30 - Lath, and Board Products, Gypsum.
TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking Of.
TT-T-291 - Thinner-paint, Volatile Spirits Petroleum Spirits.
LLL-B-1188 - Building Board, Hard Pressed, Vegetable Fiber (Laminated).

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 8010

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 562 - Consistency of Paints Using Stormer Viscosimeter.
D 1296 - Odor Test.
D 3335 - Determination of Low Concentration of Lead in Paints by Atomic Absorption Spectroscopy.
E 97 - 45-Degree, 0-Degree Directional Reflectance.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The sealer as received shall consist of the pigment and vehicle specified, so combined as to produce a sealer meeting all the requirements of this specification.

3.1.1 Pigment. The manufacturer is given wide latitude in the selection of pigments or combination of pigments provided the finished product meets all the requirements specified herein.

3.1.2 Vehicle. The vehicle shall be any suitable blend of process drying oils or varnish with drier provided the sealer meets all requirements. The solvent or thinner shall be mineral spirits conforming to TT-T-291, Type II or any solvent system which complies with Rule 66. 1/

3.2 Quantitative requirements. The sealer shall conform to the quantitative requirements of table I and 3.2.1.

TABLE I. Quantitative requirements of the sealer

Characteristics	Requirements	
	Minimum	Maximum
Nonvolatile vehicle, percent by weight of vehicle	42	---
Absorption, inch	---	1/8
Water, percent by weight of sealer	---	1.0
Consistency, Krebs-Stormer, shearing rate 200 rpm		
Grams	150	225
Equivalent Krebs Units (K.U)	70	84
Drying time:		
Set to touch, hours,	1/2	4
Dry hard or dry through for recoating, hours	---	24
Daylight directional reflectance, white only, percent	75	---
Lead content, percent by weight of total nonvolatile	---	0.5

3.2.1 Dry opacity. Not more than 7.0 ml. per sq. ft. of the white sealer, minimum apparent reflectivity of 75 percent, shall be required to give a dry-film contrast ratio of 0.84. The minimum contrast ratio for tints, applied to a wet film thickness of 1.0 ml per sq. ft., in terms of apparent reflectivity shall be as shown in table II.

TABLE II. Minimum dry-film contrast ratios for tints

Apparent reflectivity of tint	Contrast ratio	Apparent reflectivity of tint	Contrast ratio
Percent		Percent	
74	0.84	66	0.87
72	.85	64	.88
70	.85	62	.89
68	.86	60	.90

Note: A sealer having an apparent reflectivity less than 60 percent is not regarded as a tint under this specification.

1/ Information on "Rule 66" may be obtained from Air Pollution Control District County of Los Angeles, Los Angeles, CA 90013.

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3.3 Qualitative requirements.

3.3.1 Condition in container. The freshly open container when tested as in 4.3.1 shall not show evidence of caking, dry hard setting, gelling or skinning. It shall remix easily to a homogeneous state.

3.3.2 Storage stability.

3.3.2.1 Partially full container. A three-quarter filled, 1/2 pint multiple friction top can of sealer when tested as in 4.3.2.1 shall show no skinning within 48 hours. It shall show no livering, gelling, hard caking or gummy sediment.

3.3.2.2 Full container. The sealer shall be useable after 12 months storage in a standard quart can when tested as in 4.3.2.2. It shall show no caking, hard settling or gummy sediment, gelling and shall mix readily to a homogenous state. There shall be no change in the physical properties required in table I. A certificate of compliance to this requirement is acceptable.

3.3.3 Color. The color of the sealer specified in the contract or order (see 6.2) shall match the standard color in Fed. Std. No. 595, when tested as in 4.3.3.

3.3.4 Dilution stability. When tested as in 4.3.4, the sealer shall remain stable and shall show no curdling, precipitation, nor separation.

3.3.5 Odor. The odor of the sealer when tested as in 4.3.5 shall not be putrid, offensive or irritating. There shall be no residual odor after 24 hours.

3.3.6 Working properties. When brushed, rolled or sprayed on wallboard as specified in 4.3.6, the sealer shall flow and level uniformly. It shall dry to a smooth uniform finish free from brush marks, conspicous laps, streaks, etc.

3.3.7 Sealing properties (ability to prevent spotting of the top coat). When tested as in 4.3.7, the sealer shall thoroughly seal the substrate in one coat. The difference in 60° gloss between the one coat system against the two coat system shall not be over 10 percent.

3.3.8 Flexibility. A film of the sealer when tested as in 4.3.8 shall show no cracking and flaking.

3.3.9 Knife test. When tested as in 4.3.9, the sealer shall ribbon without flaking or otherwise loosening from the panel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling, inspection, and testing. Unless otherwise specified, sampling, inspection and testing of primer shall be in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.3 Test procedures. The tests shall be conducted in accordance with Fed. Test Method Std. No. 141, except as otherwise specified herein. Test methods are indicated in table III.

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TABLE III. Index

Characteristics	Requirements reference	Applicable Tests: Fed. Test Method Std. No. 141	Applicable Tests: Paragraph reference
Condition in container	3.3.1	----	4.3.1
Storage stability	3.3.2	----	4.3.2
Partially full container	3.3.2.1	3021	4.3.2.1
Full container	3.3.2.2	3022	4.3.2.2
Color	3.3.3	4250	4.3.3
Dilution stability	3.3.4	4203	4.3.4
Odor	3.3.5	----	4.3.5
Working properties	3.3.6	----	4.3.6
Sealing properties	3.3.7	----	4.3.7
Flexibility	3.3.8	6221	4.3.8
Knife test	3.3.9	6304	4.3.9
Absorption	Table I	4421	-------
Nonvolatile vehicle	Table I	4051	-------
Consistency	Table I	----	4.3.10
Drying time	Table I	4061	-------
Directional reflectance	Table I	----	4.3.11
Dry Opacity (contrast ratio)	Table II, 3.2.1	4122	-------
Lead content	Table I	----	4.3.12
Water content	Table I	4081	-------

4.3.1 Condition in container. Determine package condition of the primer in accordance with method 3011 of Fed. Test Method Std. No. 141 and evaluate for compliance with 3.3.1.

4.3.2 Storage stability.

4.3.2.1 Partially full container. Determine skinning of the sealer in a partially full container after 48 hours in accordance with method 3021 of Fed. Test Method Std. No. 141, except use a 3/4 filled 1/2-pint multiple friction top can. Reseal and store for 7 days at 140° F., then observe for compliance with 3.3.2.1.

4.3.2.2 Full container. In accordance with ASTM D 1296, allow a full standard quart can of the sealer to stand undisturbed for one year, then examine the contents for compliance with 3.3.2.2. Agitate the contents for 5 minutes then run other applicable tests. Evaluate for compliance with 3.3.2.2. A certified statement from the supplier to this effect is acceptable.

4.3.3 Color. Draw down a film on a white opaque glass panel using an applicator that will deposit a dry film thickness of approximately 1-1/2 mils. Air dry the panels at room temperature, then compare the color in accordance with method 4250 of Fed. Test Method Std. No. 141 for compliance with 3.3.3.

4.3.4 Dilution stability. Mix in accordance with method 4203 of Fed. Test Method Std. No. 141, 8 parts by volume of sealer to 1 part by volume of mineral spirits conforming to TT-T-291, type II, grade A, and observe for compliance with 3.3.4.

4.3.5 Odor. Determine the odor in accordance with ASTM D 1296 for compliance with 3.3.5. After 24 hours drying evaluate for residual odor.

4.3.6 Working properties. Prepare a 4-foot square panel of composition wallboard conforming to LLL-B-1188 or gypsum wallboard conforming to SS-L-30. Mark off the board in foot sections, starting from left as part 1; leave part 1 bare. On part 2 brush a coat of sealer at spreading rate 500 sq. ft. per gallon; on part 3 roll a coat of sealer at the same spreading rate and part 4, spray a coat at about 500 sq. ft. per gallon. After 24 hours air drying evaluate for compliance with 3.3.6.

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4.3.7 Sealing properties. Use the panel prepared in 4.3.6. Measure the gloss of the four sections, then brush a coat of the sealer on the four sections of the board, spreading of 500 sq. ft. per gallon, and allow to dry for 24 hours. Measure 60° gloss on the four sections. Compare the gloss of the one-coat section against the two-coat section and evaluate for compliance with 3.3.7.

4.3.8 Flexibility. Determine flexibility in accordance with method 6221 of Fed. Test Method Std. No. 141. Apply a 2-inch wide film with an applicator that will deposit a dry film thickness of approximately 1 mil on a tin panel (solvent clean) prepared in accordance with method 2012 of Fed. Test Method Std. No. 141. Air dry in a horizontal position at room temperature for 48 hours, then bend over 1/8-inch mandrel. Examine for compliance with 3.3.8.

4.3.9 Knife test. Cut the film from a flat portion of the panel used in 4.3.8, and examine in accordance with method 6304 of Fed. Test Method Std. No. 141 for compliance with 3.3.9.

4.3.10 Consistency. Determine consistency of the sealer in accordance with ASTM D 562, for compliance with the requirement specified in table I.

4.3.11 Directional reflectance. In accordance with ASTM E 97, run directional reflectance of the dried film for compliance with the requirement specified in table I.

4.3.12 Lead content. Determine lead content of the sealer in accordance with ASTM D 3335, for compliance with table I.

4.3.13 Inspection for preparation for delivery. The packaging, packing, and marking shall be examined and tested to determine compliance with section 5 in accordance with TT-P-143.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The sealer shall be packaged, packed and marked in accordance with TT-P-143. The level of packaging shall be A, B, or C and the level of packing shall be A, B, or C as specified (see 6.2). The sealer shall be in 1-gallon metal can, 5-gallon steel pails, or in 55-gallon steel drums as specified (see 6.2).

6. NOTES

6.1 Intended use. The sealer is an oil-type pigmented wall sealer of the "controlled penetration" type, supplied ready to use, for sizing interior wall surfaces such as plaster, wallboard, brick, unglazed tile, and concrete. Its use assures best appearance to the finished job, whether the finish coat be flat under TT-P-30; semigloss under TT-E-508; highgloss under TT-E-506 or flat latex under TT-P-29.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Administrative provisions for inspection records (see 4.1).
(c) Level of packaging and packing required (see 5.1).
(d) Size of unit container (see 5.1).
(e) Color required (see 3.3.3)

6.3 Basis of purchase. The filler should be purchased by volume, the unit being a gallon of 231 cubic inches at 15.6° C. (60° F.).

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6.4 The sealer covered by this specification is suitable for brushing or rolling application at package consistency. For spray application, instruction from the manufacturer is necessary.

Military Interest:

Navy - YD, MC

Civil Agency Coordinating Activities:

GSA-FSS
HEW-NIH
JUS-FPI
Postal-POS
VA-DMS

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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